Sahaj Bhadja

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Summary

Full-stack AI Engineer with 2+ years shipping production **LLM** workflows, **payment** integrations with idempotent webhooks, and **TypeScript**, Next.js, and Node.js apps on PostgreSQL, with multi-tenant security, failure recovery, and monitoring.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Engineer *Los Angeles, CA*

- Restructured fault-tolerant **LLM orchestration** with background workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs at 3x throughput without cascading failures under concurrent production load.
- Designed **multi-tenant** PostgreSQL and Supabase data models with row-level security for 500+ users, cutting dashboard latency from 1.2s to 350ms, and ran reconciled integrations at 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Shipped a production AI workflow that chains rubric generation, vision OCR, and grading with **structured outputs** across GPT-4o, Claude, and Qwen-VL, serving 150+ instructors and 10K+ daily requests.
- Built switchable model-provider routing across OpenAI, Anthropic, and Qwen with automatic fallbacks, timeout handling, and per-call token and **cost logging**, measuring 90%+ grading consistency while cutting instructor feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built cross-platform microservices with object-oriented Factory and Singleton patterns, covered by 20+ automated **integration tests**, and cut manual data entry 35% through automated data flows for business teams.
- Optimized third-party API integrations with pagination, field filtering, and caching, improving response times 30%, and redesigned AWS Glue and S3 pipelines for 68% higher **throughput** in Agile sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **TypeScript** Next.js and Node.js platform on PostgreSQL with JWT authentication, **role-based permissions**, and billing, invoice, and payment services, syncing 2M+ records across locations in real time.
- Monitored production with Sentry, Azure Monitor, and Kibana and gated releases on Jest and Selenium tests in CI/CD, reaching 95%+ successful build validation and cutting deployment time 50%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built multi-currency **payment infrastructure** for cards, wallets, and PayPal, with **idempotent webhooks** for refunds, disputes, and chargebacks, cutting manual refund handling about 70% across 50K+ booking and payment records.
- Built coupon validation and price recalculation with eligibility, expiry, and usage rules in booking APIs under 200ms, and kept payment status consistent with transactional locking across mobile and web.

Skills

Languages: TypeScript, JavaScript, Python, SQL, Java
AI Layer: LLM Systems, Structured Outputs, RAG, Embeddings, FAISS, Pinecone, Prompt Engineering, OCR
Full Stack: Next.js, React, Node.js, Express.js, FastAPI, REST APIs, Microservices, WebSockets
Data & Payments: PostgreSQL, Supabase, MySQL, Row-Level Security, Stripe, PayPal, Webhooks, Idempotency
Production: AWS, Vercel, Docker, GitHub Actions, CI/CD, Git, Jest, Selenium, Sentry, Kibana, Redis

Projects

Multi-Processor Payment Router | *TypeScript, NestJS, PostgreSQL, Stripe, Adyen, SQS*

- Designed a **NestJS** payment layer using a strategy pattern across Stripe and Adyen, with tokenized cards, auth-capture-refund flows, failover routing, and **webhook** events normalized into one internal model.

Merchant Operations Agent | *TypeScript, Next.js, MCP, LLM Tool Calling, PostgreSQL*

- Designed an **agent** exposing refunds, subscription changes, and order lookups as **MCP tools** with tenant-scoped authorization, idempotency keys, human approval for money movement, audit logs, and task-completion evals.

PiXMod Live | *Next.js, Stripe, CLIP, SAM, Stable Diffusion, MongoDB*

- Built an AI-powered SaaS with **Stripe** subscription billing, SAM segmentation, Stable Diffusion editing, and CLIP search behind a responsive Next.js interface, increasing premium conversion 18% for paying users.